



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO MANICKAMALAYAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

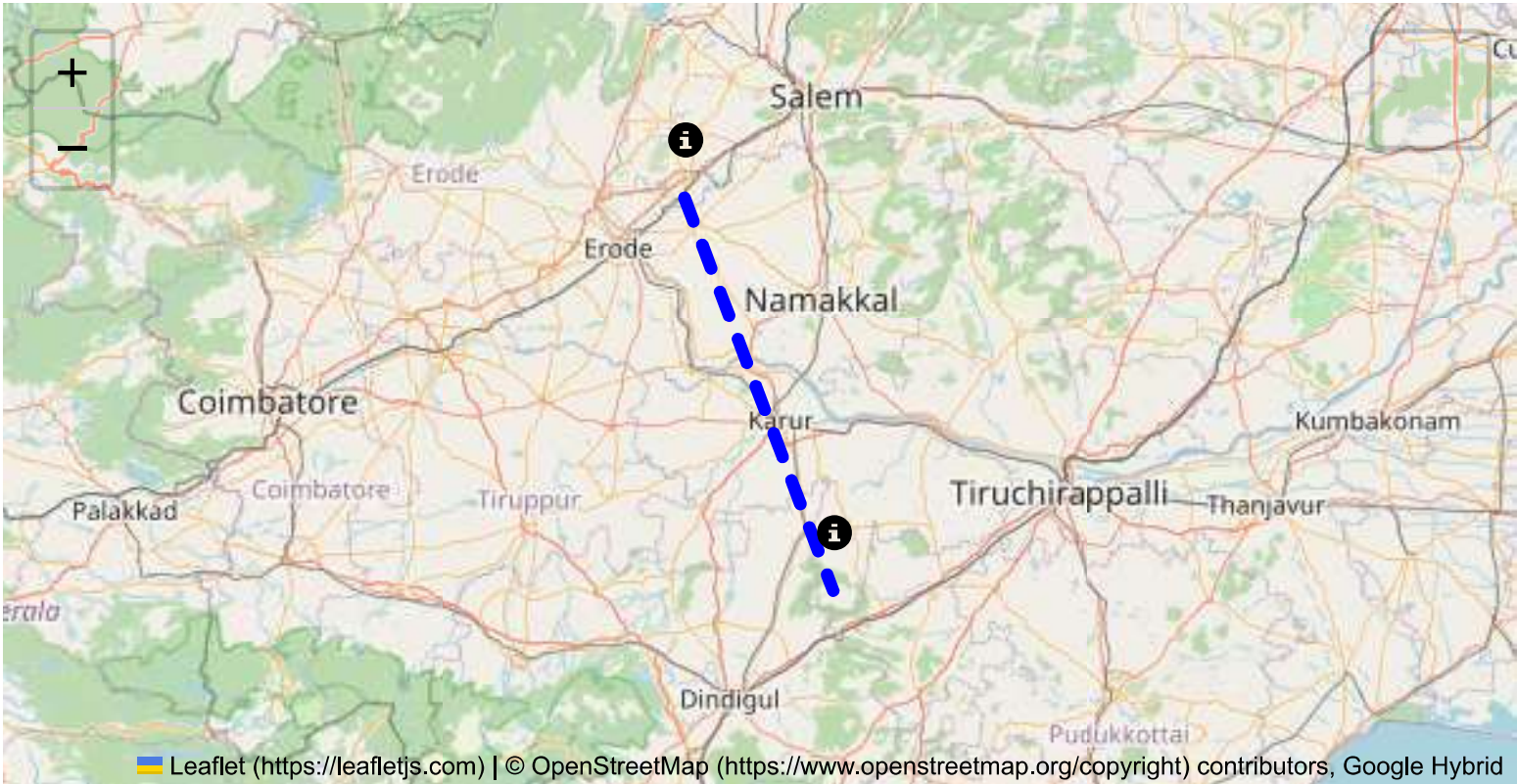
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

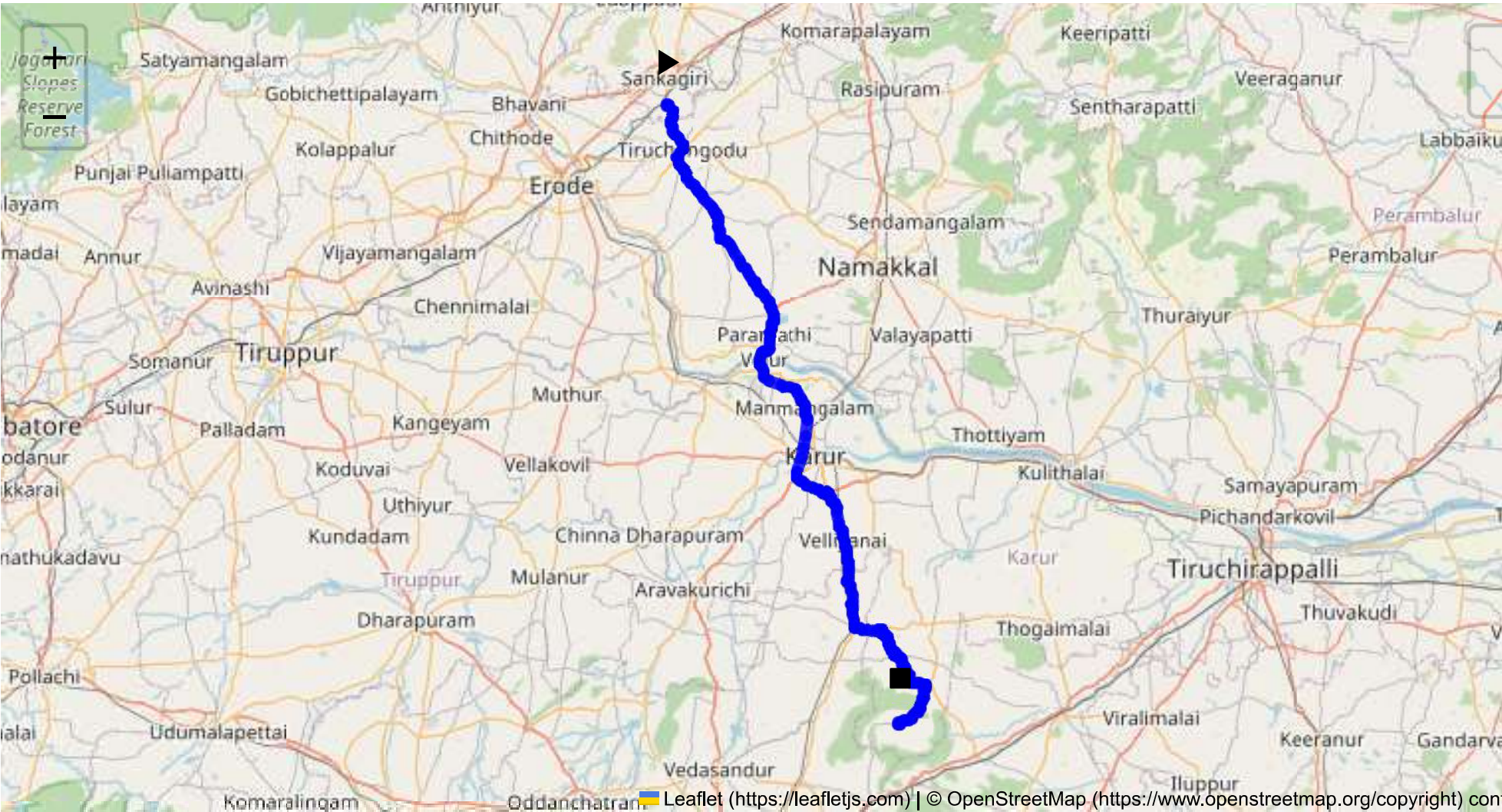
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 121.83 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (10.594718, 78.195168)



Welcome to the Journey Risk Management Study

To conduct a comprehensive analysis of the route from CVQF+23W, Sangagiri to H5WW+37F, D.Edayapatti West, this report will be broken down according to the specified criteria. The objective is to inform and train a truck driver on the potential hazards encountered along this route.

1. Overview of the Route Map:

- The route takes you through a mixture of rural and semi-urban areas in Tamil Nadu. It includes travel on national, state highways, and local roads with varying levels of infrastructure.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu generally experiences tropical weather. The region can have intense rainfall from October to December due to the northeast monsoon, leading to potential water-logged roads and reduced visibility. Summers are hot, potentially affecting vehicle cooling systems.

3. Analysis of Traffic Patterns:

- The route is likely to encounter peak traffic around major towns and road junctions, especially during morning (8-10 AM) and evening (5-7 PM) rush hours. Key congestion-prone areas include town centers and intersections with major state highways.

4. Assessment of Road Quality and Infrastructure:

- Road quality may vary. Highways are generally maintained well but local roads might have potholes, narrow segments, and inadequate signage. There may also be construction points, particularly in the monsoon season affecting travel flow.

5. Suggestions for Alternative Routes for Emergencies:

- Identify key bypass routes through smaller towns such as Namakkal, Kulithalai, and Karur. These can serve as alternate paths in case of roadblocks or accidents that obstruct the main routes.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- State regulations mandate permits for transporting hazardous materials. Specific route permissions may be required, and adherence to axle load limits is critical.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- While specific details are scarce, typical incidents may include overturns and spills due to sharp turns or sudden braking. Areas near Sangagiri and Namakkal are noted for such occurrences due to high traffic density and steep inclines.

8. Environmental Considerations and Sensitive Areas:

- Numerous agricultural zones lie along the route, requiring cautious material handling to avoid contamination. Some sections might pass near water bodies; spill prevention is crucial.

9. Analysis of Communication Coverage:

- Cellular coverage is robust in urban and semi-urban areas but might be patchy in rural stretches. Notable dead zones could be in the hilly or forested areas near Anai Mudi.

10. Estimated Emergency Response Times:

- Urban centers such as Sangagiri and Karur generally have quicker emergency services (15-30 minutes). However, rural areas may experience delays, with response times extending to 45 minutes or more.

11. Overall Summary of Risk Assessment:

- The route poses moderate risk, primarily due to weather conditions, variable road quality, and traffic congestion. Weather patterns necessitate planning for monsoon disruptions. Proper vehicle maintenance is crucial to minimize breakdown risks in remote areas. Understanding

regulations and emergency procedures is vital for transporting hazardous materials safely. By using alternative routes and timing journeys to avoid peak traffic, risks can be mitigated effectively. Regular updates from local traffic and weather services are recommended for real-time decision-making.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	U-Turn	High	10.9110477, 78.09713769999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	Medium	11.37902, 77.89500	30 KM/Hr
7	Turn	Medium	11.37850, 77.89453	30 KM/Hr
8	Turn	Medium	11.16950, 78.01544	30 KM/Hr
9	Turn	Medium	11.16764, 78.01565	30 KM/Hr
10	Turn	Medium	11.15546, 78.02170	30 KM/Hr
11	Turn	High	11.14521, 78.02239	15 KM/Hr
12	Turn	High	11.14480, 78.02291	15 KM/Hr
13	Turn	Medium	11.10102, 78.00572	30 KM/Hr
14	Turn	Medium	11.08294, 78.00335	30 KM/Hr
15	Turn	Medium	11.05261, 78.05138	30 KM/Hr
16	Turn	Medium	10.92908, 78.05505	30 KM/Hr
17	Turn	Medium	10.90753, 78.09394	30 KM/Hr
18	Turn	Medium	10.90730, 78.09763	30 KM/Hr
19	Blind Spot	Blind Spot	10.90793, 78.09916	10 KM/Hr
20	Blind Spot	Blind Spot	10.91105, 78.09714	10 KM/Hr
21	Turn	Medium	10.81788, 78.12357	30 KM/Hr
22	Turn	Medium	10.78879, 78.12489	30 KM/Hr
23	Turn	Medium	10.75989, 78.13139	30 KM/Hr
24	Turn	Medium	10.75772, 78.13010	30 KM/Hr
25	Turn	Medium	10.75558, 78.13104	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
26	Turn	Medium	10.72618, 78.13583	30 KM/Hr
27	Turn	High	10.72580, 78.13601	15 KM/Hr
28	Turn	Medium	10.72241, 78.14604	30 KM/Hr
29	Turn	High	10.72145, 78.16655	15 KM/Hr
30	Turn	High	10.72346, 78.16703	15 KM/Hr
31	Turn	High	10.72265, 78.17354	15 KM/Hr
32	Turn	Medium	10.71884, 78.17426	30 KM/Hr
33	Blind Spot	Blind Spot	10.71810, 78.17408	10 KM/Hr
34	Turn	Medium	10.71800, 78.17522	30 KM/Hr
35	Turn	Medium	10.71743, 78.17572	30 KM/Hr
36	Turn	Medium	10.71421, 78.17604	30 KM/Hr
37	Turn	Medium	10.70289, 78.18316	30 KM/Hr
38	Turn	Medium	10.70200, 78.18437	30 KM/Hr
39	Turn	High	10.69823, 78.18475	15 KM/Hr
40	Turn	High	10.69780, 78.18539	15 KM/Hr
41	Turn	High	10.69403, 78.18652	15 KM/Hr
42	Turn	High	10.69421, 78.18730	15 KM/Hr
43	Turn	Medium	10.69314, 78.18822	30 KM/Hr
44	Turn	High	10.69197, 78.18847	15 KM/Hr
45	Turn	High	10.69146, 78.19202	15 KM/Hr
46	Turn	Medium	10.69027, 78.19205	30 KM/Hr
47	Turn	Medium	10.68708, 78.19063	30 KM/Hr
48	Turn	High	10.68669, 78.19068	15 KM/Hr
49	Turn	Medium	10.68560, 78.19717	30 KM/Hr
50	Turn	High	10.68312, 78.20051	15 KM/Hr
51	Turn	High	10.68024, 78.20065	15 KM/Hr
52	Turn	High	10.67972, 78.20149	15 KM/Hr
53	Turn	Medium	10.67622, 78.20086	30 KM/Hr
54	Turn	High	10.66152, 78.21040	15 KM/Hr
55	Turn	High	10.65166, 78.20938	15 KM/Hr
56	Turn	High	10.64662, 78.23481	15 KM/Hr
57	Turn	Medium	10.60662, 78.21679	30 KM/Hr
58	Turn	Medium	10.60270, 78.21527	30 KM/Hr
59	Turn	High	10.60079, 78.21365	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
60	Blind Spot	Blind Spot	10.59513, 78.19657	10 KM/Hr
61	Turn	High	10.59482, 78.19694	15 KM/Hr
62	Blind Spot	Blind Spot	10.59402, 78.19691	10 KM/Hr
63	Turn	High	10.59417, 78.19543	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9110477, 78.09713769999999



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37902, 77.89500



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16950, 78.01544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15546, 78.02170



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14521, 78.02239



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08294, 78.00335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05261, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92908, 78.05505



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90753, 78.09394



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90730, 78.09763



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.90793, 78.09916



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.91105, 78.09714



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.81788, 78.12357



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.78879, 78.12489



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.75989, 78.13139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75772, 78.13010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75558, 78.13104



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.72618, 78.13583



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.72580, 78.13601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.72241, 78.14604



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72145, 78.16655



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72346, 78.16703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72265, 78.17354



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71884, 78.17426



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.71810, 78.17408



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71800, 78.17522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71743, 78.17572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71421, 78.17604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70289, 78.18316



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70200, 78.18437



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69823, 78.18475



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69780, 78.18539



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69403, 78.18652



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69421, 78.18730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.69314, 78.18822



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69197, 78.18847



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.69146, 78.19202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.69027, 78.19205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.68708, 78.19063



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.68669, 78.19068



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.68560, 78.19717



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.68312, 78.20051



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.68024, 78.20065



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.67972, 78.20149



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.67622, 78.20086



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.66152, 78.21040



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.65166, 78.20938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.64662, 78.23481



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.60662, 78.21679



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.60270, 78.21527



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.60079, 78.21365



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.59513, 78.19657

Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.59482, 78.19694
